

Question 1:

A standard deck has:

- 4 Kings (one in each suit)
- 13 Hearts (one in each rank)
- 1 King of Hearts (the card that is both a King and a Heart)

The events of drawing a King and drawing a Heart are not mutually exclusive because the King of Hearts is common to both. Therefore, we use the General Rule of Addition:

$$P(K \text{ or } H) = P(K) + P(H) - P(K \text{ and } H)$$

Now,

$$P(K) = \frac{\text{Number of Kings}}{\text{Total Cards}} = \frac{4}{52}$$

$$P(H) = \frac{\text{Number of Hearts}}{\text{Total Cards}} = \frac{13}{52}$$

$$P(K \text{ and } H) = \frac{\text{Number of King of Hearts}}{\text{Total Cards}} = \frac{1}{52}$$

$$\therefore P(K \text{ or } H) = P(K) + P(H) - P(K \text{ and } H)$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$= \frac{16}{52}$$

$$= \frac{4}{13}$$

Question 2:

The probability that an American Automobile Association (AAA) member made airline reservations last year is $P(A) = 60\%$ or 0.60 . We are selecting two members at random, and we want the probability that both made airline reservations. Since the members are selected randomly and independently of each other, we use the Multiplication Rule for Independent Events:

$$\begin{aligned} P(M_1 \text{ and } M_2) &= P(M_1) \times P(M_2) \\ &= 0.60 \times 0.60 \\ &= 0.36 \end{aligned}$$

$\therefore$ The probability that both members made airline reservations last year is 0.36

Question 3:

A golfer has a total of 12 golf shirts.

- White shirts (W): 9
- Blue shirts (B): $12 - 9 = 3$

The golfer selects two shirts in a row without replacement (does not do laundry). We want the probability that both shirts are white. This requires the General Rule of Multiplication, as the second selection depends on the first (dependent events):

$$P(W_1 \text{ and } W_2) = P(W_1) \times P(W_2 | W_1)$$

1. Probability the first shirt is white, $P(W_1)$:

$$\begin{aligned} P(W_1) &= \frac{\text{Number of White Shirts}}{\text{Total Shirts}} \\ &= \frac{9}{12} \end{aligned}$$

2. Probability the second shirt is white given the first was white, $P(W_2 | W_1)$:

After the first white shirt is selected, there are now 8 white shirts left and 11 total shirts left.

$$\begin{aligned} P(W_2 | W_1) &= \frac{\text{White Shirts Remaining}}{\text{Total White Shirts}} \\ &= \frac{8}{11} \end{aligned}$$

$$\begin{aligned} \therefore P(W_1 \text{ and } W_2) &= P(W_1) \times P(W_2 | W_1) \\ &= \frac{9}{12} \times \frac{8}{11} = \frac{72}{132} \sim 0.55 \end{aligned}$$

The likelihood both shirts selected are white is $72/132$ or approximately 0.55

Question 4:

The number of executives in the cell A_1 and B_4 is 75. The total number of executives surveyed is 200

$$\therefore P(A_1 \text{ and } B_4) = \frac{\text{Number of Executive in } A_1 \text{ and } B_4}{\text{Total Number of Executives}} = \frac{75}{200} = 0.375$$

Question 5:

A and B mutually exclusive, given, $P(A) = 0.30$, $P(B) = 0.20$

$$\therefore P(A \text{ or } B) = P(A) + P(B)$$

$$= 0.30 + 0.20$$

$$= 0.50$$

$$\therefore P(\text{neither}) = 1 - 0.50 = 0.50$$

Question 6:

$$\text{a) } P(I_1) = \frac{\text{Number of Firms} < \$1 \text{ Million}}{\text{Total Number of Firms}} = \frac{102}{200} = 0.51$$

b) Let, probability of selecting a firm with income between \$1 million and \$20 million is $P(I_2)$ probability of selecting a firm with income \$20 million is $P(I_3)$

$$\therefore P(I_2 \text{ or } I_3) = P(I_2) + P(I_3) = \frac{61}{200} + \frac{37}{200} = \frac{49}{100} = 0.49$$

Question 7:

The event of getting a grade above a C includes getting an A or getting a B. Since a student can only get one final grade, the events A and B are mutually exclusive.

We use the Special Rule of Addition:

$$P(\text{Above C}) = P(A \text{ or } B) = P(A) + P(B) = 0.25 + 0.50 = 0.75$$

Question 8:

Given, $P(A) = 0.20$, $P(B) = 0.30$ and $P(A \text{ and } B) = 0.15$

We want the probability of either A or B occurring, $P(A \text{ or } B)$. We use the General Rule of Addition:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = 0.20 + 0.30 - 0.15 = 0.35$$

Question 9:

Two events A and B are mutually exclusive if the occurrence of one event means the other event cannot occur.

The probability of their joint occurrence, $P(A \text{ and } B)$, is the probability that both events happen at the same time.

If A and B are mutually exclusive, $P(A \text{ and } B) = 0$.

Answer: The probability of their joint occurrence is 0.

Question 10:

Let P be a pharmacy, F be a floral shop, and D be a deli.

Given,

- $P(P) = 0.40$
- $P(F) = 0.50$
- $P(D) = 0.70$
- $P(P \text{ and } F \text{ and } D) = 0.10$ (All three)
- $P(P \text{ and } D) = 0.30$ (Pharmacy and Deli)
- $P(F \text{ and } D) = 0.25$ (Floral Shop and Deli)
- $P(P \text{ and } F) = 0.20$ (Pharmacy and Floral Shop)

a) $P(P \text{ and } F) = 0.20$

b) $P(P \text{ and } D) = 0.30$

c) Two events are mutually exclusive if $P(D \text{ and } P) = 0$. We are given $P(P \text{ and } D) = 0.30$. Since $0.30 \neq 0$, the events are not mutually exclusive

d) The name for “selecting a store with pharmacy, floral, and deli” is the **joint event** (written $P(P \cap F \cap D)$).

e) The event "does not have all three" is the **complement** of the event "has all three" $P(P \text{ and } F \text{ and } D)$

$$\begin{aligned} P(\text{Not All Three}) &= 1 - P(P \text{ and } F \text{ and } D) \\ &= 1 - 0.10 \\ &= 0.90 \end{aligned}$$

Probability the store has at least one of the three departments is:

$$\begin{aligned} P(P \cup F \cup D) &= P(P) + P(D) + P(F) - P(P \text{ and } D) - P(F \text{ and } D) - P(P \text{ and } F) \\ &= 0.40 + 0.50 + 0.70 - (0.30 + 0.20 + 0.25) + 0.10 \\ &= 0.95 \end{aligned}$$

Question 11:

Given, $P(A) = 0.40$, $P(B|A) = 0.30$

Joint probability:

$$\begin{aligned}P(A \cap B) &= P(A) \times P(B|A) \\&= 0.40 \times 0.30 \\&= 0.12\end{aligned}$$

Question 12:

Let C be checking account and S be savings account. Given,

$P(C) = 0.80$, $P(S) = 0.60$ and $P(C \text{ and } S) = 0.50$

$$\therefore P(C \text{ or } S) = P(C) + P(S) - P(C \text{ and } S)$$

$$= 0.80 + 0.60 - 0.50$$

$$= 0.90$$

The event "does not have either" is the complement of the event "has either a checking or a savings account."

$$P(\text{Neither}) = 1 - P(C \text{ or } S)$$

$$= 1 - 0.90 = 0.10$$

Question 13:

a) Given, Total for $A_1 = 3$ and Grand Total = 10

$$\therefore P(A_1) = \frac{\text{Total for } A_1}{\text{Grand Total}} = \frac{3}{10} = 0.30$$

b) $P(B_1 | A_2)$ is the conditional probability of event B_1 occurring given that event A_2 has already occurred.

We look only at the A_2 column (the condition) and divide the joint count of (A_2 and B_1) by the total count for A_2

- Count for (A_2 and B_1) = 1
- Total for $A_2 = 3$

$$\text{We know, } P(B_1 | A_2) = \frac{(B_1 \text{ and } A_2)}{A_2} = \frac{1}{3} = 0.33$$

c) $P(B_2 \text{ and } A_3)$ is the joint probability of both events B_2 and A_3 occurring. It is calculated by dividing the count of the cell where B_2 and A_3 intersect by the grand total.

- Count for $(B_2 \text{ and } A_3) = 1$
- Grand Total = 10

$$\therefore P(B_2 \text{ and } A_3) = \frac{B_2 \text{ and } A_3}{\text{Grand Total}} = \frac{1}{10} = 0.1$$

Question 14:

- a) A table that cross-classifies data for two variables is called a Contingency Table
b) The number of salespeople in the cell S_3 and E is 135. The total number of salespeople is 500

$$P(S_3 \text{ and } E) = \frac{\text{Number of Salespeople with } S_3 \text{ and } E}{\text{Total Number of Salespeople}} = \frac{135}{500} = 0.27$$

Question 15:

There are 5 board members. Strong Leaders (L) = 3 and Weak Members (W) = 2

Probability the First Presentation Wins (S_1):

The first salesperson wins if they randomly select one of the 3 strong leaders out of the 5 board members.

$$\therefore P(S_1) = P(L_1) = \frac{3}{5} = 0.60$$

Probability the Second Presentation Wins (S_2):

For Second to win, two things must happen in order:

1. First must fail (pick a weak). Probability First fails = $\frac{2}{5}$.
After First picks a weak, remaining board members = 4 total: 3 strong, 1 weak.
2. Given First failed, Second must pick a strong from the 4 remaining: probability $\frac{3}{4}$

Multiply conditional probabilities:

$$P(\text{Second wins}) = P(\text{First fails}) \times P(\text{Second picks S} \mid \text{First failed}) = \frac{2}{5} \times \frac{3}{4} = \frac{6}{20} = 0.30$$

Probability the Third Presentation Wins (S_3):

For Third to win, both First and Second must fail (pick the two weak members), then Third is guaranteed to pick a strong (because only the 3 strong remain).

- First fails: probability $\frac{2}{5}$.
- After First fails, one weak and 3 strong remain (4 total). Second must also fail by picking the remaining weak: probability $\frac{1}{4}$.
- If both failed, Third then has only 3 strong left out of 3. Picks strong with probability 1.

Multiply:

$$P(\text{Third wins}) = \frac{2}{5} \times \frac{1}{4} \times 1$$

$$= \frac{2}{20}$$

$$= 0.10$$